

Nathan Machuki Onduma

Senior MEL and Data Science Leader

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Professional Summary

Senior Monitoring, Evaluation, and Learning (MEL) and applied data science leader with 15+ years of experience across global health and development programs in Africa and international settings. Proven track record leading multi-country evidence, learning, and performance systems that support strategic decision-making and program improvement. Expertise in statistical analysis, program evaluation, implementation learning, and translating complex data into actionable insights for governments, donors, and implementing partners. Experienced in strengthening health information systems, managing large-scale programs and partnerships, and supporting evidence-informed strategy.

Core Competencies

- Strategic Measurement, Evaluation, and Learning (MEL)
- Evidence Systems, Adaptive Learning, and Decision Support
- Quantitative & Qualitative Analysis and Applied Data Science
- Program Design, Evaluation Frameworks, and Implementation Learning
- Global Health Information Systems (HMIS, DHIS2, EMR, LMIS, eCHIS, KoboToolbox, ODK)
- Performance Measurement, Analytics, and Organizational Learning
- Multi-country Program Performance and Partner Engagement
- Stakeholder Engagement and Capacity Strengthening
- Strategic Donor Reporting (USAID, CDC, World Bank, Foundations, and Funds)
- Research, Evidence Generation, and Scientific Communication

Professional Experience

The END Fund

Senior Associate Director, Measurement, Evaluation, and Learning

March 2025 – Present / Remote, Kenya

- Design and lead MEL strategy and implementation for six NTD programs within multi-country initiatives spanning 30+ countries, supporting over \$50M in disbursements and reaching 150M+ individuals annually.
- Oversee development of MEL frameworks, indicators, and data systems to strengthen program performance and impact measurement.
- Provide technical leadership in survey design, data analytics, and digital transformation initiatives.
- Support organizational strategy through advanced data analysis and evidence generation for donor and board reporting.
- Deliver technical assistance to Ministries of Health and implementing partners on data demand and use for decision-making, automated reporting, DHIS2 workflows, and surveillance analytics.
- Provide data-driven technical support for fund development, including guidance during the proposal development stage.

The END Fund

Associate Director, Measurement, Evaluation, and Learning

October 2022 – February 2025 / Remote, Kenya

- Designed and implemented MEL systems across 30+ low-resource countries with limited M&E capacity, strengthening data use for programmatic decision-making.
- Led digitization initiatives across Nigeria and DRC, enhancing data collection and reporting for five PC NTD programs.
- Supported integration of visceral leishmaniasis indicators into national HMIS platforms.
- Contributed to major investment cases and strategic documents that informed fundraising and program prioritization.
- Strengthened partner accountability via performance tracking and data reviews, aligning country MEL systems with global MEL frameworks and donor reporting standards.
- Developed and managed quick-use data tools, including Google Sheets and Excel dashboards, to support program teams.

Selected Achievements

- Advanced multi-country data systems, including indicator frameworks, data tools, shapefiles, and training resources.
- Delivered high-impact analytics supporting localization strategy and cross-departmental decision-making.
- Contributed to development of organizational Theory of Change aligned with strategic priorities.
- Led segments of the DHIS2-based program database development in collaboration with [BAO systems](#), the technical contractor.

Independent Consultant

Statistician / MEL Specialist

October 2020 – October 2022

- Delivered statistical and MEL consulting services for global health and development programs funded by USAID, World Bank, and other donors and partners.
- Designed and implemented data analysis frameworks and evaluation methodologies.
- Translated complex analytical outputs into clear, actionable insights for technical and non-technical stakeholders.
- Provided advisory support on monitoring, measurement, evaluation, research, and learning for public, private, and non-profit organizations.
- Provided technical leadership on planning and conducting a [Harmonized Health Facility Assessment \(HHFA\)](#) and third party monitoring

University of Maryland Global Initiatives Corporation

Strategic Information Advisor

February 2017 – October 2020 / Kenya, PACT Timiza Program

- Led MEL strategy and implementation for a USG PEPFAR/CDC-funded HIV program.
- Directed data systems including DATIM, DHIS2, and EMR platforms to ensure accurate and timely reporting.
- Managed a multidisciplinary team of MEL and data professionals.
- Provided technical assistance to county health systems to strengthen data collection, analysis, and reporting.
- Led baseline assessments, data quality assessments, routine reporting, and stakeholder review meetings.

Selected Achievements

- Increased EMR system coverage from below 20% to over 70%.
- Owned the development and deployment of the DHIS2-based Data Reporting and Warehousing System (DRAWS) with [BAO systems](#), the technical contractor.
- Part of the technical working group for the development and deployment of CDC Kenya's Partner Platform for Performance Monitoring (3PM).
- Designed and implemented a comprehensive Evaluation and Performance Monitoring Plan.
- Secured continued donor funding through high-quality reporting and performance tracking.
- Supervised 2 data managers, 7 M&E officers, and 1 HMIS specialist.

KMET

Monitoring and Evaluation Officer

April 2016 – January 2017 / Kisumu, Kenya

- Developed and managed monitoring and evaluation systems for donor-funded RMNCAH (Reproductive, Maternal, Newborn, Child, and Adolescent Health) programs.
- Developed indicators, data collection tools, reporting matrices, and institutional data systems.
- Led training and capacity building for program staff, service delivery staff, and partners.
- Supported proposal development, grant reporting, data quality audits, and research dissemination.

Selected Achievements

- Improved DHIS2 reporting quality and coverage from 55% to 94% within five months.
- Recognized as top-performing partner in reporting quality and timeliness across 8 implementing partners.
- Supervised 2 M&E assistants directly and 30 data clerks indirectly.

Ministry of Planning, National Development and Vision 2030

Economist/Statistician

September 2010 – April 2016 / Kenya

- Led county-level statistical analysis, planning, monitoring, and reporting functions.
- Developed program monitoring tools, performance indicators, and policy inputs.
- Coordinated development, review, monitoring, and evaluation of District Development Plans and County Integrated Development Plans.
- Supported budgeting, strategic planning, feasibility studies, and sector statistics dissemination.

Selected Achievements

- Led the production of [Nyamira County's inaugural Statistical Abstract](#) in collaboration with the Kenya National Bureau of Statistics providing a foundation for evidence-based economic planning.
- Led the development of [the inaugural County Integrated Development Plan](#), multi-year (2013/2014 to 2015/2016) budget policy documents, and strategic frameworks, plans and [reports](#), ensuring alignment with county and national priorities.
- Developed Nyamira County M&E indicators in partnership with UNDP and UN-Women to support tracking of the key performance indicators.
- Founded the Borabu District Information and Documentation Centre, improving transparency and accessibility of government development plans and strategies for civil servants and citizens.

Education

- [Master of Science in Statistics](#), Kenyatta University, Kenya — 2014
- [Bachelor of Science in Statistics, Second Class Upper](#), University of Nairobi, Kenya — 2009
- [Kenya Certificate of Secondary Education, A Plain](#), Nyamagwa Boys' High School, Kenya — 2003

Technical Skills

- **Programming & Analysis:** R tidyverse ecosystem, Excel/Google Sheets, Power BI
- **Health Systems:** DHIS2, EMR systems, Kobo Toolbox
- **Enterprise Systems:** Salesforce
- **Research & Reporting:** Survey design, statistical modelling, data visualization, LaTeX

Selected Professional Development

- **Time Series Forecasting** — [Prof. Bahman Rostami-Tabar](#), Nairobi, Kenya, 2023
- **Demographic and Health Surveys (DHS) Program online courses**, 2022
- **Leadership and Management in Health** — University of Washington, 2020
- **Introduction to Epidemiology for Global Health** — University of Washington, 2018
- **Monitoring and Evaluation fundamentals, frameworks, data management, and analysis** — CDC Kenya, Ministry of Health, Global Health Learning Center, and related partners
- **Spatial Statistics** — [Prof. Thomas Achia](#), CDC Kenya, 2017

Selected Publications

- Duncan Ochol, Kebron Haile, **Nathan Onduma**, Tsegaye Yohanes, & Kimberly Kamara (2025). *Elimination of visceral leishmaniasis in Ethiopia: cross-sector collaboration and cost sharing to promote sustainability*. [PubMed](#)
- Makau-Barasa, L. K., **Onduma, N.**, Yotebieng, K., & Karutu, C. (2023). *African institutions will lead on the road to end neglected tropical diseases*. [Frontiers](#)
- **Onduma, N. M.** (2019–2022). *RPubs publications and technical notes on statistical and data science methodologies*. [RPubs](#).
- **Onduma, N. M.** (2014). *Mixture regression-cum-ratio estimators using multi-auxiliary variables and attributes in two-phase sampling*. MSc project, Kenyatta University.
- Mshambala, A. W., **Onduma, N. M.**, & Ng'ang'a, J. G. (2009). *Online presence of internet consumers in Nairobi's Central Business District*. Undergraduate project, University of Nairobi.

Languages

- English — Full professional proficiency
- Kiswahili — Full professional proficiency
- French — Basic proficiency

Selected Awards

- **2019:** Recognized by the University of Maryland Kenya Country Director as “*First Among Equals*” for leadership of the Strategic Information Department.
- **2007:** Awarded university fees waiver after being recognized as the Best Second-Year Student, School of Mathematics, University of Nairobi.
- **2006:** Best First-Year Student, Statistics Section, University of Nairobi.
- **2003:** Ranked second-best KCSE student in Gucha District, Kenya.

Referees

Available upon request.